

CO2-AT-Part2

Application-Based Question

Digital Forensics Using TD(0), SARSA, and Q-Learning

1. TD(0) Learning

Pseudocode

Initialize state value $V(s)$ for all forensic investigation states.

Set learning rate α .

Set discount factor γ .

Observe the current investigation state.

Repeat until the investigation is completed

 Perform forensic investigation action.

 Receive immediate reward.

 Observe the next investigation state.

 Update the state value using TD(0).

 Move to the next state.

End Repeat

Return the optimal state values.

Proposed Algorithm for TD(0)

Step 1: Initialize Investigation States

Initialize the forensic investigation states and assign an initial value function.

$$V(s) = 0 \quad (1)$$

where

$V(s)$ = Initial value of investigation state.

Eq. (1) initializes every investigation state with zero prior knowledge before learning begins.

Step 2: Collect Digital Evidence

For every investigation state

- System Log Analysis
- Network Packet Inspection
- Malware Detection
- File Recovery
- Reward Collection

The agent receives an immediate reward after performing the investigation.

Step 3: Compute Temporal Difference Error

$$\delta = r_{t+1} + \gamma V(s_{t+1}) - V(s_t) \quad (2)$$

where

r_{t+1} = Immediate reward

γ = Discount factor

$V(s_{t+1})$ = Next state value

$V(s_t)$ = Current state value

Eq. (2) computes the Temporal Difference error by comparing the observed reward with the previously estimated investigation value.

Step 4: Update Investigation State

$$V(s_t) = V(s_t) + \alpha \delta \quad (3)$$

where

α = Learning rate

δ = TD error from Eq. (2)

Eq. (3) updates the value of the current investigation state using the TD error obtained from Eq. (2), enabling the agent to improve future forensic decisions.

Step 5: Output

Return the optimal investigation state values.

The state having the maximum value is selected as the best forensic investigation path.

Calculation and Prediction

Given

$$\alpha = 0.2 \quad (4)$$

$$\gamma = 0.9 \quad (5)$$

$$V(s_t) = 25 \quad (6)$$

$$r_{t+1} = 12 \quad (7)$$

$$V(s_{t+1}) = 30 \quad (8)$$

Step 1: Compute TD Error

Using Eq. (2)

$$\delta = 12 + 0.9(30) - 25 \quad (9)$$

$$\delta = 12 + 27 - 25 = 14 \quad (10)$$

Step 2: Update State Value

Using Eq. (3)

$$V(s) = 25 + 0.2(14) \quad (11)$$

$$V(s) = 27.8 \quad (12)$$

Prediction

The updated investigation value is **27.8**. Since the value increased after learning, the TD(0) agent predicts that the current investigation path is effective for identifying digital evidence and recommends continuing this forensic investigation strategy.

2. SARSA

Pseudocode

Initialize $Q(s,a)$ for all state-action pairs.

Set learning rate α .

Set discount factor γ .

Set exploration rate ϵ .

Observe the current investigation state.

Choose an action using ϵ -greedy policy.

Repeat until the investigation is completed

Perform the selected investigation action.

Receive immediate reward.

Observe the next investigation state.

Choose the next action using ϵ -greedy policy.

Update the Q-value using SARSA.

Move to the next state and action.

End Repeat

Return the optimal Q-values.

Proposed Algorithm for SARSA

Step 1: Initialize State-Action Values

Initialize the Q-value for every forensic investigation state and action.

$$Q(s, a) = 0 \quad (13)$$

where

$Q(s, a)$ = Initial state-action value.

Eq. (13) initializes all state-action pairs with zero values before learning starts.

Step 2: Select Investigation Action

Choose the current forensic investigation action using the ϵ -greedy policy.

$$a = \epsilon\text{-Greedy}(Q) \quad (14)$$

where

a = Selected investigation action

ϵ = Exploration probability

Eq. (14) balances exploration of new investigation paths and exploitation of previously learned forensic knowledge.

Step 3: Update the State-Action Value

$$Q(s, a) = Q(s, a) + \alpha[r + \gamma Q(s', a') - Q(s, a)] \quad (15)$$

where

$Q(s, a)$ = Current Q-value

r = Immediate reward

γ = Discount factor

$Q(s', a')$ = Next state-action value

α = Learning rate

Eq. (15) updates the current state-action value using the reward received and the next action selected by the current policy.

Step 4: Output

Return the optimal state-action values.

The investigation action having the highest Q-value is selected for digital forensic analysis.

Calculation and Prediction

Given

$$\alpha = 0.2 \quad (16)$$

$$\gamma = 0.9 \quad (17)$$

$$Q(s, a) = 40 \quad (18)$$

$$r = 15 \quad (19)$$

$$Q(s', a') = 45 \quad (20)$$

Step 1: Compute Updated Q-value

Using **Eq. (15)**

$$Q(s, a) = 40 + 0.2[15 + 0.9(45) - 40] \quad (21)$$

$$Q(s, a) = 40 + 0.2(15 + 40.5 - 40) \quad (22)$$

$$Q(s, a) = 40 + 0.2(15.5) \quad (23)$$

$$Q(s, a) = 40 + 3.1 \quad (24)$$

$$Q(s, a) = 43.1 \quad (25)$$

Prediction

The updated Q-value is **43.1**. Since the value has increased from **40** to **43.1**, the SARSA agent predicts that the selected sequence of forensic actions is effective for evidence collection and should continue following the current investigation policy.

3. Q-Learning

Pseudocode

Initialize $Q(s,a)$ for all state-action pairs.

Set learning rate α .

Set discount factor γ .

Set exploration rate ϵ .

Observe the current investigation state.

Repeat until the investigation is completed

 Choose an investigation action using ϵ -greedy policy.

 Perform the selected investigation action.

 Receive immediate reward.

 Observe the next investigation state.

 Find the maximum Q-value of the next state.

 Update the current Q-value.

 Move to the next investigation state.

End Repeat

Return the optimal Q-values.

Proposed Algorithm for Q-Learning

Step 1: Initialize State-Action Values

Initialize the Q-value for every forensic investigation state and action.

$$Q(s, a) = 0 \quad (26)$$

where

$Q(s, a)$ = Initial state-action value.

Eq. (26) initializes all state-action values with zero before the learning process begins.

Step 2: Select Investigation Action

Choose the forensic investigation action using the ϵ -greedy policy.

$$a = \epsilon\text{-Greedy}(Q) \quad (27)$$

where

a = Selected investigation action

ϵ = Exploration probability

Eq. (27) selects either the best-known investigation action or explores a new action to improve future learning.

Step 3: Update State-Action Value

$$Q(s, a) = Q(s, a) + \alpha \left[r + \gamma \max_{a'} Q(s', a') - Q(s, a) \right] \quad (28)$$

where

$Q(s, a)$ = Current Q-value

r = Immediate reward

γ = Discount factor

$\max_{a'} Q(s', a')$ = Maximum Q-value of the next state

α = Learning rate

Eq. (28) updates the Q-value using the maximum possible future reward, allowing the agent to learn the optimal investigation policy independently of the action actually taken.

Step 4: Output

Return the optimal state-action values.

The investigation action with the highest Q-value is selected as the optimal digital forensic strategy.

Calculation and Prediction

Given

$$\alpha = 0.2 \quad (29)$$

$$\gamma = 0.9 \quad (30)$$

$$Q(s, a) = 40 \quad (31)$$

$$r = 15 \quad (32)$$

$$\max_{a'} Q(s', a') = 50 \quad (33)$$

Step 1: Compute Updated Q-value

Using **Eq. (28)**,

$$Q(s, a) = 40 + 0.2[15 + 0.9(50) - 40] \quad (34)$$

$$Q(s, a) = 40 + 0.2(15 + 45 - 40) \quad (35)$$

$$Q(s, a) = 40 + 0.2(20) \quad (36)$$

$$Q(s, a) = 40 + 4 \quad (37)$$

$$Q(s, a) = 44 \quad (38)$$

Prediction

The updated Q-value is **44**. Since the Q-value has increased from **40** to **44**, the Q-Learning agent predicts that the selected investigation action is optimal for maximizing future rewards. Therefore, the digital forensic system recommends following this investigation strategy to improve evidence collection and cyber threat detection.